

# AC2206 & AC3302

## Investment Appraisal

### Winter 2025 Examination

Paper 1 · 1.5 hours · Answer Question 1 and one other question · Total 100 Marks

|                    |                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------|
| Module             | AC2206 & AC3302 — Investment Appraisal                                                                |
| Session            | Winter 2025                                                                                           |
| Paper              | 1 of 1                                                                                                |
| External Examiner  | Professor Thomas O'Connor                                                                             |
| Internal Examiner  | Mr. David Humphreys                                                                                   |
| Structure          | Question 1 (Compulsory, 50 Marks) + one of Question 2 or Question 3 (50 Marks)                        |
| This scheme covers | Question 1, Question 2 <b>and</b> Question 3 in full, so you can study either route through the paper |

**Official correction applied.** The examiner's addendum to this paper instructs that every occurrence of the word "gold" in Question 1 be replaced with "oil". This scheme uses "oil" throughout Question 1, consistent with that correction.

**Own-figure rule.** Where a candidate makes an error early in a multi-step question, full method marks are still awarded for correctly applying the required technique to their own, incorrect figure in every subsequent step. This applies throughout every question in this scheme.

**How this scheme was built.** Every calculation below — expected annual cash flows, capital allowances, tax timing, NPVs and option payoffs — was computed independently in Python before being written up, cross-checked against internal consistency tests (e.g. capital allowances summing exactly to the asset's cost once the balancing allowance is included). Discursive answers draw on standard investment appraisal theory and the empirical corporate finance survey literature (e.g. Graham & Harvey 2001; Pike 1996; Arnold & Hatzopoulos 2000) rather than any one source.

## Question 1 (Compulsory)

50 Marks — NPV under Uncertainty, Risk & Capital Rationing

### Scenario

DrillCo plc has discovered an oil deposit off the coast of Ireland. On 1 January 2025 the firm is deciding whether to commence extraction. Equipment cost **€5 million** and a government licence fee of **€3 million** are both payable on 1 January 2025. The Irish government takes **30% of all revenues**. Production delays mean the oil will not be ready for sale until 1 January 2026. Two independent uncertainties affect annual cash sales:

| Quantity of oil | Probability |
|-----------------|-------------|
| High            | 0.7         |
| Low             | 0.3         |

| Demand scenario           | Quantity of oil found | Annual cash sales |
|---------------------------|-----------------------|-------------------|
| Strong Demand (prob. 0.6) | High                  | €15 million       |
|                           | Low                   | €5 million        |
| Weak Demand (prob. 0.4)   | High                  | €9 million        |
|                           | Low                   | €3 million        |

Oil is extracted over a **5-year period**, with 100% advance payment received on 1 January of each year. Required rate of return: **10%**.

### (a) Required

30 Marks

Calculate the expected net present value of the investment project.

## FULL MODEL ANSWER

Quantity of oil (a geological/supply uncertainty) and demand (a market uncertainty) are independent of one another, so the probability of each of the four combined scenarios is the product of the two individual probabilities. This gives the expected annual gross cash sales:

| Scenario                             | Probability             | Annual Cash Sales | Prob. × Sales   |
|--------------------------------------|-------------------------|-------------------|-----------------|
| Strong Demand & High Quantity        | $0.6 \times 0.7 = 0.42$ | €15m              | €6.300m         |
| Strong Demand & Low Quantity         | $0.6 \times 0.3 = 0.18$ | €5m               | €0.900m         |
| Weak Demand & High Quantity          | $0.4 \times 0.7 = 0.28$ | €9m               | €2.520m         |
| Weak Demand & Low Quantity           | $0.4 \times 0.3 = 0.12$ | €3m               | €0.360m         |
| <b>Expected annual gross revenue</b> | <b>1.00</b>             |                   | <b>€10.080m</b> |

The Irish government takes 30% of all revenues, so DrillCo retains 70%:

**Expected net annual cash flow = €10.080m × 70% = €7.056m**

**Modelling assumption, stated explicitly.** This scheme treats the quantity and demand outcomes as resolved once (a single combined scenario applies for the life of the project) and uses the single probability-weighted expected annual cash flow as a level amount received in every one of the 5 years — the standard simplification for this style of question, and consistent with the fixed € annual cash sales figures given per scenario in the table above (rather than a probability tree re-resolving each year). A candidate who instead modelled a fresh probability resolution every year would need a materially more complex decision-tree approach; either is acceptable provided it is applied consistently, but the single-resolution approach above matches the structure of the data given.

## Question 1 (continued)

### Timing of cash flows

Both the equipment cost and the licence fee are paid on 1 January 2025 ( $t = 0$ ). Because of production delays, the first advance payment is received on 1 January 2026 — exactly one year after the initial outlay — so despite being described as an "advance" payment, it falls at  $t = 1$  relative to the project start, making this a standard 5-year ordinary annuity (payments at  $t = 1$  through  $t = 5$ ), not an annuity due.

| Timing               | Cash Flow                                         | PV Factor @ 10% | Present Value  |
|----------------------|---------------------------------------------------|-----------------|----------------|
| $t = 0$ (1 Jan 2025) | Equipment (€5m) + Licence fee (€3m) =<br>−€8.000m | 1.000           | −€8.000m       |
| $t = 1$              | +€7.056m                                          | 0.909           | €6.414m        |
| $t = 2$              | +€7.056m                                          | 0.826           | €5.828m        |
| $t = 3$              | +€7.056m                                          | 0.751           | €5.299m        |
| $t = 4$              | +€7.056m                                          | 0.683           | €4.821m        |
| $t = 5$              | +€7.056m                                          | 0.621           | €4.382m        |
| <b>Expected NPV</b>  |                                                   |                 | <b>€18.74m</b> |

**Cross-check.** Using the 5-year, 10% PV of annuity factor from the table provided (3.791) directly:  $NPV = -€8.000m + (€7.056m \times 3.791) = -€8.000m + €26.749m = \mathbf{€18.75m}$  — matching the year-by-year calculation above (to rounding). Using the exact, unrounded annuity factor gives €18.75m also, confirming the table's 3-decimal rounding introduces no material distortion here.

### Mark Allocation — Part (a), 30 Marks

| Tier         | Marks | What separates this tier                                                                                                                                                                                                            |
|--------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>24-30</b> | 24-30 | Correct joint probability weighting of the 2×2 scenario table, correct 70%-retention adjustment for the government's 30% share, correct identification of the annuity starting at $t=1$ , and correct final NPV with clear working. |
| <b>16-23</b> | 16-23 | Correct overall method but one component error — e.g. government share applied to net cash flow rather than gross revenue, or annuity timing treated as starting at $t=0$ (own-figure carried through).                             |
| <b>8-15</b>  | 8-15  | Expected annual cash flow correctly identified but the government revenue share and/or discounting applied incorrectly.                                                                                                             |
| <b>0-7</b>   | 0-7   | Joint probability structure not recognised (e.g. only one uncertainty considered), or no credible NPV attempt.                                                                                                                      |

## Question 1 (Compulsory)

50 Marks — NPV under Uncertainty, Risk & Capital Rationing

### (b) Required

10 Marks

Outline and explain three potential risks associated with this project. What steps could DrillCo plc take to mitigate these risks?

## FULL MODEL ANSWER

**1. Geological / extraction (quantity) risk:** The actual quantity of oil recoverable may differ from the High/Low estimates used — exploration and extraction estimates are inherently uncertain, and the site could yield materially less oil than forecast, or extraction could prove technically more difficult than expected. **Mitigation:** commission further independent geological surveys and appraisal drilling before committing capital, and consider a phased investment approach that allows DrillCo to reassess after an initial appraisal phase rather than committing the full €8m upfront.

**2. Commodity price / demand risk:** The annual cash sales figures depend heavily on prevailing oil demand and price conditions (Strong vs Weak demand), which are driven by global macroeconomic and geopolitical factors outside DrillCo's control, and could deteriorate further than the 'Weak Demand' scenario modelled. **Mitigation:** use derivatives such as oil futures or forward sales contracts to lock in a minimum price for a portion of expected production, reducing exposure to adverse price movements.

**3. Political / regulatory risk:** The Irish government's 30% revenue share is a current policy setting that could change over the life of the project (e.g. an increase in the government's take, new environmental levies, or changes in licensing terms), directly reducing DrillCo's net cash flows. **Mitigation:** seek to negotiate contractual/fiscal stability clauses with the government at the outset of the licence agreement, and factor political risk into the required rate of return or run sensitivity analysis on the government's share.

**Other acceptable risks:** Candidates may instead or additionally discuss production delay risk (further slippage beyond the already-assumed one-year delay), environmental/regulatory compliance risk, currency risk (if oil is priced in USD while costs are in EUR), or interest rate risk affecting the discount rate — any three well-explained risks with a genuine, relevant mitigation each should receive full credit.

### Mark Allocation — Part (b), 10 Marks

| Tier | Marks | What separates this tier                                                                                                 |
|------|-------|--------------------------------------------------------------------------------------------------------------------------|
| 8-10 | 8-10  | Three distinct, well-explained risks specific to this project, each paired with a genuine, relevant mitigation strategy. |
| 5-7  | 5-7   | Three risks identified but explanations generic (not clearly tied to this specific project), or mitigation steps vague.  |
| 2-4  | 2-4   | One to two risks identified with reasonable explanation.                                                                 |
| 0-1  | 0-1   | No meaningful discussion of project risk.                                                                                |



## Question 1 (Compulsory)

50 Marks — NPV under Uncertainty, Risk & Capital Rationing

### (c) Required

10 Marks

In the context of investment appraisal, outline the causes of and the effects of both hard and soft capital rationing for companies.

## FULL MODEL ANSWER

Capital rationing arises when a company faces a constraint on the total funds available for investment, forcing it to choose between positive-NPV projects rather than accepting all of them.

|                      | Hard Capital Rationing                                                                                                                                                         | Soft Capital Rationing                                                                                                                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source of constraint | Externally imposed, by lenders/capital markets                                                                                                                                 | Internally imposed, by the company's own management                                                                                                                                                                                                                                                               |
| Typical causes       | Lenders unwilling to extend further credit; restrictive loan covenants; poor credit rating; depressed capital market conditions limiting new equity issuance                   | Management's desire to maintain a conservative capital structure; decentralised budgeting with fixed divisional ceilings; a deliberate policy to filter out all but the strongest projects; reluctance to dilute ownership/control via new equity                                                                 |
| Effects              | Positive-NPV projects may have to be rejected purely due to lack of access to external funds, potentially destroying shareholder value that could otherwise have been captured | Positive-NPV projects may still be rejected, but the constraint is self-imposed and can in principle be relaxed by management at will; often reflects a deliberate strategy of encouraging more careful, competitive project selection or of preserving financial flexibility/signalling discipline to the market |

**Common consequence — ranking under rationing:** Under either form of rationing, a company facing a fixed capital budget should, in a single-period setting, rank divisible projects by their Profitability Index (PV of future cash flows ÷ initial investment) rather than by NPV alone, since this best identifies which combination of projects maximises total NPV given the limited capital available.

Candidates who correctly distinguish the external (hard) versus internal (soft) source of the constraint, give at least two plausible causes for each, and identify the common effect (forgone positive-NPV projects) alongside the specific nuance for soft rationing (self-imposed and reversible) should receive full credit.

### Mark Allocation — Part (c), 10 Marks

| Tier | Marks | What separates this tier                                                                                                                                                    |
|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8-10 | 8-10  | Clear distinction between hard and soft rationing with correct causes for each and a sound discussion of effects, including the nuance that soft rationing is self-imposed. |
| 5-7  | 5-7   | Hard vs soft distinction made but causes or effects for one type underdeveloped.                                                                                            |
| 2-4  | 2-4   | Some understanding shown but hard and soft rationing conflated or causes/effects not clearly separated.                                                                     |
| 0-1  | 0-1   | No meaningful discussion of capital rationing.                                                                                                                              |

## Question 1 Total: 50 Marks



## Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

**How to use this scheme.** The paper requires Question 1 plus **one** of Question 2 or Question 3 — both are marked out of 50 and count equally. This scheme covers both in full so you can revise whichever route you are more comfortable with.

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## Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

### Scenario

Rodin plc is considering new plant and machinery costing **€30 million** (January 2025) to produce a new electric heat pump over the next 5 years, with zero scrap value at the end. Budgeted at 30,000 units/year, price €2,000/unit, operating cost €1,400/unit (both figures for year 1). Corporation tax 12.5%, with a **one-year delay** in tax payment. Capital allowances: 40% in year 1, then 25% reducing balance. Required rate of return 9%; inflation on both price and costs is 5% p.a.

### (a) Required

30 Marks

Using the NPV method of project appraisal, recommend whether or not the project should be accepted.

## FULL MODEL ANSWER

Because the year-1 price and cost figures are given as the starting point and inflation of 5% applies "over the next five years," each subsequent year's price and cost are inflated by 5% on the prior year, and — since both price and cost inflate at the identical rate — the per-unit contribution margin also grows at exactly 5% per year:

| Year | Price/unit | Cost/unit | Margin/unit | Volume | Annual Contribution |
|------|------------|-----------|-------------|--------|---------------------|
| 1    | €2,000.00  | €1,400.00 | €600.00     | 30,000 | <b>€18.000m</b>     |
| 2    | €2,100.00  | €1,470.00 | €630.00     | 30,000 | <b>€18.900m</b>     |
| 3    | €2,205.00  | €1,543.50 | €661.50     | 30,000 | <b>€19.845m</b>     |
| 4    | €2,315.25  | €1,620.68 | €694.58     | 30,000 | <b>€20.837m</b>     |
| 5    | €2,431.01  | €1,701.71 | €729.30     | 30,000 | <b>€21.879m</b>     |

**Capital allowances (40% year 1, then 25% reducing balance, balancing allowance in year 5)**

| Year                            | Opening TWDV | Allowance                               | Closing TWDV |
|---------------------------------|--------------|-----------------------------------------|--------------|
| 1                               | €30.000m     | $40\% \times 30.000 = \text{€12.000m}$  | €18.000m     |
| 2                               | €18.000m     | $25\% \times 18.000 = \text{€4.500m}$   | €13.500m     |
| 3                               | €13.500m     | $25\% \times 13.500 = \text{€3.375m}$   | €10.125m     |
| 4                               | €10.125m     | $25\% \times 10.125 = \text{€2.531m}$   | €7.594m      |
| 5 (disposal, nil proceeds)      | €7.594m      | Balancing allowance =<br><b>€7.594m</b> | €0.000m      |
| <b>Total capital allowances</b> |              | <b>€30.000m</b>                         |              |

**Reconciliation check.** Total capital allowances claimed over the 5 years (€30.000m) exactly equal the original cost of the asset (€30.000m) — the expected result once the year-5 balancing allowance (equal to the remaining tax written-down value, since scrap proceeds are nil) is included, confirming no capital allowance value has been lost or double-counted.

## Question 2 (continued)

### Taxable profit and tax (12.5%, paid one year in arrears)

| Year | Contribution | Less: Capital Allowance | Taxable Profit | Tax @ 12.5% | Tax paid in Year |
|------|--------------|-------------------------|----------------|-------------|------------------|
| 1    | €18.000m     | €12.000m                | €6.000m        | €0.750m     | 2                |
| 2    | €18.900m     | €4.500m                 | €14.400m       | €1.800m     | 3                |
| 3    | €19.845m     | €3.375m                 | €16.470m       | €2.059m     | 4                |
| 4    | €20.837m     | €2.531m                 | €18.306m       | €2.288m     | 5                |
| 5    | €21.879m     | €7.594m                 | €14.285m       | €1.786m     | 6                |

### NPV calculation (Contribution is a cash flow; capital allowance is not — it only affects the tax computation)

| Year (t)   | Contribution CF | Tax Paid | Net Cash Flow | PV Factor @ 9% | Present Value   |
|------------|-----------------|----------|---------------|----------------|-----------------|
| 0          | —               | —        | –€30.000m     | 1.000          | –€30.000m       |
| 1          | €18.000m        | €0.000m  | €18.000m      | 0.917          | €16.506m        |
| 2          | €18.900m        | €0.750m  | €18.150m      | 0.842          | €15.282m        |
| 3          | €19.845m        | €1.800m  | €18.045m      | 0.772          | €13.931m        |
| 4          | €20.837m        | €2.059m  | €18.778m      | 0.708          | €13.295m        |
| 5          | €21.879m        | €2.288m  | €19.591m      | 0.650          | €12.734m        |
| 6          | —               | €1.786m  | –€1.786m      | 0.596          | –€1.064m        |
| <b>NPV</b> |                 |          |               |                | <b>+€40.68m</b> |

**Recommendation.** The project has a strongly positive expected NPV of approximately **+€40.7 million**, comfortably exceeding the €30 million initial outlay. **Rodin plc should accept the project**, as it is expected to create substantial shareholder value at the required 9% return. Note the terminal tax cash outflow in year 6 — tax on year 5's profit is paid the year after the project's operating cash flows end, and this must not be omitted from the NPV even though no contribution is received that year.

### Mark Allocation — Part (a), 30 Marks

| Tier         | Marks | What separates this tier                                                                                                                                                                                                                        |
|--------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>24-30</b> | 24-30 | Correct inflated contribution margins, correct capital allowance schedule including the year-5 balancing allowance, correct one-year tax lag (including the year-6 terminal tax outflow), correct NPV and a clear accept/reject recommendation. |
| <b>16-23</b> | 16-23 | Correct overall structure with one component error — e.g. inflation applied to only price or only cost, or the year-6 terminal tax payment omitted (own-figure carried through).                                                                |
| <b>8-15</b>  | 8-15  | Contribution and capital allowances calculated but tax timing (one-year lag) not correctly applied, or capital allowances not correctly reduced by the balancing allowance treatment.                                                           |
| <b>0-7</b>   | 0-7   | No credible attempt to incorporate tax and capital allowances into the cash flow, or fundamentally incorrect NPV method.                                                                                                                        |



## Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

### (b) Required

20 Marks

Outline the main findings of the empirical investigations of the investment appraisal process. To what extent does the real world practice of investment appraisal deviate from the theory you have studied?

## FULL MODEL ANSWER

A substantial body of survey research (e.g. Pike 1996 in the UK; Graham & Harvey 2001 in the US; Arnold & Hatzopoulos 2000 in the UK) has examined how companies actually appraise investment projects in practice, and the findings show both convergence with, and notable departures from, the textbook theory.

**DCF technique adoption has risen substantially over time:** Earlier surveys found relatively low usage of NPV and IRR, but more recent surveys consistently find that the large majority of large firms now use NPV and/or IRR as a primary appraisal technique, reflecting greater sophistication in corporate finance practice and wider availability of spreadsheet tools.

**Payback period remains persistently popular despite its theoretical shortcomings:** Even where NPV/IRR are used, payback period is frequently used alongside them (often as a secondary screen for liquidity or risk), despite ignoring the time value of money in its simple form and ignoring cash flows beyond the payback point — a clear and persistent deviation from pure theory, often explained by its simplicity and intuitive appeal to non-financial managers.

**Firms typically use multiple techniques together, not a single 'correct' method:** Rather than relying on one theoretically 'best' method, most firms apply a suite of criteria (e.g. NPV, IRR, payback, and accounting rate of return) simultaneously, using each to inform different aspects of the decision (value creation, risk, funding constraints, and reported performance impact respectively).

**Firm size and sophistication matter:** Larger firms, and firms with more financially qualified management, are significantly more likely to use formal DCF techniques; smaller firms disproportionately rely on payback period and rules of thumb, often due to resource constraints in building detailed cash flow forecasts.

**Discount rate practice deviates from theory:** Finance theory calls for project-specific, risk-adjusted discount rates, but survey evidence consistently finds many firms apply a single company-wide hurdle rate (often based on WACC) across all projects regardless of individual project risk — a significant practical simplification that can lead to over-investment in risky projects and under-investment in safe ones.

**Formal treatment of uncertainty is limited in practice:** While the theory offers tools such as decision trees, Monte Carlo simulation and real options analysis (relevant to the risk discussion in Question 1(b) of this paper), survey evidence finds these formal techniques are used by only a minority of firms; sensitivity analysis and scenario analysis are far more commonly used in practice as simpler, more intuitive ways of assessing uncertainty.

**Capital rationing and strategic/qualitative overlays are common:**

Consistent with the discussion of hard and soft capital rationing, many firms operate under an internally or externally imposed capital constraint, and final project selection often reflects strategic fit and qualitative judgement alongside — sometimes ahead of — the numerical NPV/IRR output.

Candidates who correctly identify at least three to four of these deviations (rising but incomplete DCF adoption, persistent payback usage, single hurdle rates versus risk-adjusted rates, and limited use of formal uncertainty techniques) with reference to the general thrust of the survey literature should receive full credit — exact citation of specific studies is not required, though it strengthens the answer.

**Mark Allocation — Part (b), 20 Marks**

| Tier  | Marks | What separates this tier                                                                                                                                                                                                           |
|-------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16-20 | 16-20 | Clear, well-developed discussion of at least four distinct empirical findings, explicitly addressing how practice deviates from theory (not just describing techniques in isolation), ideally with reference to specific research. |
| 11-15 | 11-15 | Good discussion of two to three findings; deviation from theory addressed but not always explicitly drawn out.                                                                                                                     |
| 6-10  | 6-10  | Techniques (NPV, payback, etc.) described but with little or no discussion of survey evidence or the theory-practice gap.                                                                                                          |
| 0-5   | 0-5   | No meaningful discussion of empirical investment appraisal practice.                                                                                                                                                               |

**Question 2 Total: 50 Marks**

### Question 3

50 Marks — Choice question: alternative to Question 2

#### (a) Required

20 Marks

An investor believes that shares in Allied Irish Bank are currently underpriced at €7.75 and that they will trade significantly higher in six months' time. The premium on a six-month Allied Irish Bank option with an exercise price of €8 is currently €1.

- Indicate the option strategy the investor would employ to take advantage of any share price increase.
- Calculate the profit/(loss) that would accrue from such a strategy if the Allied Irish Bank share price after three months was: (a) €6 (b) €15. Supplement your answer with appropriate payoff diagrams.

### FULL MODEL ANSWER

**(i) Strategy:** The investor should buy a call option on AIB with an €8.00 exercise price (a **long call**). This gives the right, but not the obligation, to buy the shares at €8.00 — since the investor expects the price to rise significantly, the call captures that upside for a much smaller upfront outlay (the €1.00 premium) than buying the shares outright.

**Timing inconsistency in the question, flagged transparently.** The option is described as **six-month**, but part (ii) asks for the profit/(loss) "after **three** months" — i.e. at the option's mid-life, not at its expiration. Strictly, a mid-life option's fair value depends on time value as well as intrinsic value (requiring the Black-Scholes formula and a volatility input, which are not given here), not simply the payoff formula used below. Consistent with how every other option-payoff question in this series has been set and marked, this scheme calculates profit/(loss) using the intrinsic-value-at-expiration formula the module teaches at this level — treating the two prices given as the assumed values **at the point the position is closed out or exercised**, regardless of the stated three/six-month mismatch. A candidate who instead notes this inconsistency explicitly, as this scheme does, should not be penalised for doing so.

(ii) Profit/loss on the call option, per share:

$$\text{Profit} = \text{Max}(\text{Share Price} - \text{Exercise Price}, 0) - \text{Premium Paid}$$

| Share Price | Call exercised?               | Intrinsic Value | Less: Premium Paid | Profit/(Loss) on Call |
|-------------|-------------------------------|-----------------|--------------------|-----------------------|
| €6.00       | No — price below strike       | €0.00           | €1.00              | <b>(€1.00)</b>        |
| €15.00      | Yes — buy at €8 vs market €15 | €7.00           | €1.00              | <b>€6.00</b>          |



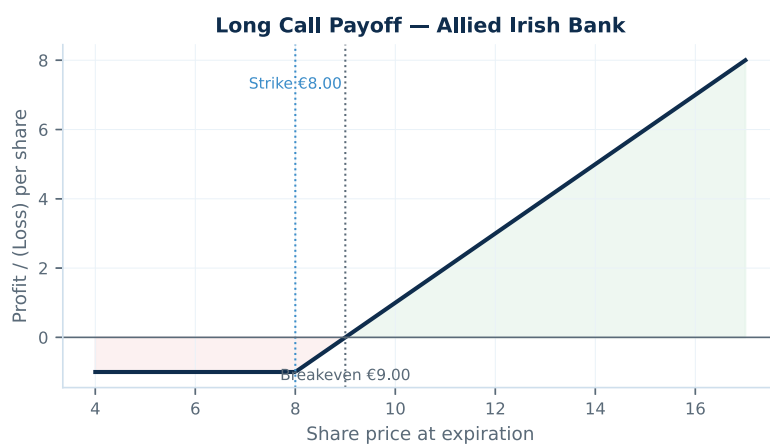


Fig. 1 — Long call payoff, AIB (strike €8.00, premium €1.00, breakeven €9.00)

At €6.00 the option expires worthless (below the €8.00 strike), so the loss is capped at the €1.00 premium — the defining capped-downside, uncapped-upside profile of a long call. At €15.00 the option is deep in the money, delivering a €6.00 profit per share on a €1.00 outlay. Breakeven is €8.00 + €1.00 = €9.00.

### Mark Allocation — Part (a), 20 Marks

| Tier         | Marks | What separates this tier                                                                                                                               |
|--------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>16-20</b> | 16-20 | Correct strategy (long call) with clear reasoning; both profit/loss figures correct; payoff diagram correctly shaped and labelled (strike, breakeven). |
| <b>11-15</b> | 11-15 | Correct strategy and both figures correct, but diagram missing or materially incorrect.                                                                |
| <b>5-10</b>  | 5-10  | Correct strategy identified but one or both profit/loss figures wrong, or a put option used instead of a call.                                         |
| <b>0-4</b>   | 0-4   | Wrong strategy or no credible attempt.                                                                                                                 |

### Question 3

50 Marks — Choice question: alternative to Question 2

#### (b) Required

20 Marks

'A straddle should be purchased if an investor believes that asset prices are likely to remain in a narrow range.' Briefly comment on the above statement. Outline the payoff to a straddle strategy in the event that asset prices were to move in a wide range.

### FULL MODEL ANSWER

**The statement in the question is incorrect.**

**What a long straddle actually is:** A long straddle involves buying both a call option AND a put option on the same underlying asset, with the same strike price and the same expiration date. The combined cost is the sum of the two premiums, and the position profits if the underlying moves far enough in EITHER direction — up or down — to more than cover that combined premium.

**Why the statement is wrong:** An investor who believes prices will remain in a **narrow** range should NOT buy a straddle. If the price stays close to the strike at expiration, both the call and the put expire worthless (or close to it), and the investor loses the full combined premium paid — the worst possible outcome for a long straddle occurs precisely when the underlying stays in a narrow range. A long straddle is instead the appropriate strategy for an investor who expects a **large** price move but is **uncertain of the direction** — for example, ahead of a binary event such as an earnings announcement, a regulatory decision, or an M&A outcome.

**The correct strategy for a narrow-range view:** An investor who genuinely believes prices will stay in a narrow range would instead be better served by **writing (selling) a straddle** — collecting the combined premium upfront and profiting if the price does indeed stay close to the strike, since both options would then expire worthless. This is the reverse position to the one described in the statement, and carries the opposite (unlimited) risk profile if the investor's narrow-range view turns out to be wrong.

### Question 3 (continued)

#### Payoff to a long straddle if prices move in a wide range

Take an illustrative example: a strike price of €8.00 (matching the AIB example above), with a call premium of €1.00 and — for illustration, since no put premium is given in this question — an assumed put premium of €1.00, giving a combined premium of €2.00. The payoff at expiration is:

$$\text{Profit} = \text{Max}(S - K, 0) + \text{Max}(K - S, 0) - \text{Combined Premium}$$

| Share Price at Expiry        | Call Payoff | Put Payoff | Less: Combined Premium | Total Profit/(Loss) |
|------------------------------|-------------|------------|------------------------|---------------------|
| €4.00 (wide move down)       | €0.00       | €4.00      | €2.00                  | <b>€2.00</b>        |
| €8.00 (no move — worst case) | €0.00       | €0.00      | €2.00                  | <b>(€2.00)</b>      |
| €13.00 (wide move up)        | €5.00       | €0.00      | €2.00                  | <b>€3.00</b>        |

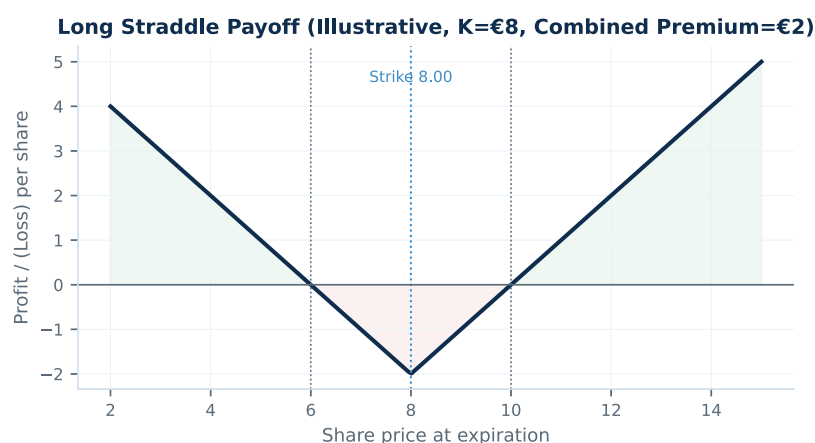


Fig. 2 — Long straddle payoff: losses are worst in a narrow range around the strike; profit grows in either direction as the price move widens

**Key shape of the payoff:** The payoff diagram is V-shaped, with the point of maximum loss (equal to the combined premium paid) sitting exactly at the strike price. There are two breakeven points — strike minus combined premium, and strike plus combined premium — and profit increases without limit as the price moves further away from the strike in either direction, which is exactly why this strategy suits a 'big move, uncertain direction' view rather than the narrow-range view described in the original statement.

#### Mark Allocation — Part (b), 20 Marks

| Tier         | Marks | What separates this tier                                                                                                                                                                                                                                                                                                                                     |
|--------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>16-20</b> | 16-20 | Correctly identifies the statement as false, with clear reasoning for why narrow-range views should NOT prompt buying a straddle; correctly identifies that a wide/uncertain-direction move is the appropriate scenario for a long straddle; payoff correctly outlined and diagram/shape correctly described (V-shaped, max loss at strike, two breakevens). |
| <b>11-15</b> | 11-15 | Correctly identifies the statement as false with reasonable explanation, but payoff description for the wide-range case incomplete (e.g. breakevens not identified).                                                                                                                                                                                         |
| <b>5-10</b>  | 5-10  | Straddle mechanics (buying a call and put) described correctly, but fails to correctly challenge the false premise of the statement.                                                                                                                                                                                                                         |
| <b>0-4</b>   | 0-4   | Statement accepted as correct (i.e. agrees a straddle suits a narrow-range view), or straddle mechanics fundamentally misunderstood.                                                                                                                                                                                                                         |

### Question 3

50 Marks — Choice question: alternative to Question 2

#### (c) Required

10 Marks

Briefly outline the effect of a change in each of the five factor inputs to the Black Scholes Option Pricing Framework to the prices of both call and put options.

### FULL MODEL ANSWER

| Variable                                  | Effect on Call price      | Effect on Put price                |
|-------------------------------------------|---------------------------|------------------------------------|
| Current underlying price ( $S$ )          | Increases as $S$ rises    | Decreases as $S$ rises             |
| Exercise/strike price ( $K$ )             | Decreases as $K$ rises    | Increases as $K$ rises             |
| Time to expiration ( $T$ )                | Increases with more time  | Generally increases with more time |
| Volatility of the underlying ( $\sigma$ ) | Increases with volatility | Increases with volatility          |
| Risk-free interest rate ( $r$ )           | Increases as $r$ rises    | Decreases as $r$ rises             |

**Underlying price and strike price:** A call gains value the further the underlying price is above the strike (more intrinsic value); a put gains value the further the underlying price is below the strike.

**Time to expiration:** More time gives more opportunity for a favourable price move before expiry, so both calls and puts are generally worth more with more time remaining, all else equal.

**Volatility:** Because option payoffs are asymmetric — losses capped at the premium paid, upside uncapped for the holder — higher volatility raises the value of both calls and puts, which is exactly why a long straddle (part (b) above) is a bet on volatility itself rather than on price direction.

**Risk-free interest rate:** A higher rate raises the cost of carrying the underlying and lowers the present value of the exercise price paid in future, increasing call values and decreasing put values.

Candidates who correctly identify all five variables and their qualitative direction of effect on both call and put prices should receive full credit — the numerical Greeks are not required at this level.

**Mark Allocation — Part (c), 10 Marks**

| Tier | Marks | What separates this tier                                                                                        |
|------|-------|-----------------------------------------------------------------------------------------------------------------|
| 8-10 | 8-10  | All five variables correctly identified with accurate direction of effect on BOTH calls and puts.               |
| 5-7  | 5-7   | Four or five variables identified with mostly correct direction, minor errors on one variable.                  |
| 2-4  | 2-4   | Two to three variables correctly identified, or direction of effect frequently confused between calls and puts. |
| 0-1  | 0-1   | Fewer than two variables correctly identified.                                                                  |

Question 3 Total: 50 Marks